



COLLEGE OF COMPUTER AND INFORMATION SCIENCE

Academic Year 2024-2025

FINAL REPORT

Reiner Gabrielle Requina VALDEZ

Submitted to the faculty of Malayan Colleges Laguna
In Partial Fulfillment of the Requirements for the Degree of

Bachelor of Science in Information Technology

OVERVIEW OF THE PRACTICUM ENGAGEMENT

COMPANY BACKGROUND



Figure 1. iNeed Solutions Inc. Logo

iNeed Solution Inc., operating as iNeed Parking founded by Jonarc Cardines on year 2023. It was born from the universal frustration of circling city blocks in search of available parking spaces after experiencing firsthand the stress of navigating unfamiliar streets and settling for distant parking spots. The company is dedicated to transforming the urban parking experience through innovative technology. By connecting drivers with available parking spaces seamlessly, iNeed Parking aims to eliminate the time-consuming search for parking and reduce the stress associated with finding convenient spots in busy city areas. The company's goal is to make city parking convenient and accessible for everyone, turning the simple phrase "iNeed Parking" into an effortless solution that guides drivers directly to their ideal parking spot.

COMPANY MISSION & VISION

- MISSION

Connect the parkers to thousands of available and accessible parking spaces anytime and anywhere - all online.

- VISION

Transforming all non-performing land space becomes valuable, useful, and efficiently managed for the customers, owners, cities, and on a national scale.

NATURE OF ASSIGNMENT OR TASK GIVEN

During my practicum, she was assigned to a variety of tasks that utilized diverse tools and techniques. Each participant completed 2 to 3 individual tasks weekly, resulting in a total of 24 tasks. These tasks involved using tools such as Visual Studio Community and Android Studio, along with technologies like MVC, SignalR, Flutter, and MongoDB. Currently, she is contributing to the development of a mobile parking application that connects parkers with parking space owners.

ROLE

Junior Software Engineer

TOTAL HOURS RENDERED

To complete the 486 hours requirement for the Practicum of the course Information Technology, 8 Hours were allotted to Course orientation, consisting of an introduction, explaining the basics, plans, and the process of communication. While 382 Hours were given to software development, including the individual tasks, team tasks, planning, and designing. For the Career Planning section, 80 hours were given, and 16 hours for documentation, which includes organization, planning, logging, video submission, and any written documentation.

```

1  using System;
2  namespace HelloWorld
3  {
4      class Program
5      {
6          static void Main()
7          {
8              Console.WriteLine("Hello World!");
9          }
10     }
11 }

```

As shown in Figure 3, the Task Scheduler was utilized to create an automatic logging per hour starting from a time. The output was put in the .txt file that has a format of '[date] [time]' written at the date and time, and to create an automated task scheduler that will send a message through email per hour using our Practicum email and our email. Both automations were created through Visual Studio Community, then were put on the Task Scheduler, setting it up and readying it for the automation.

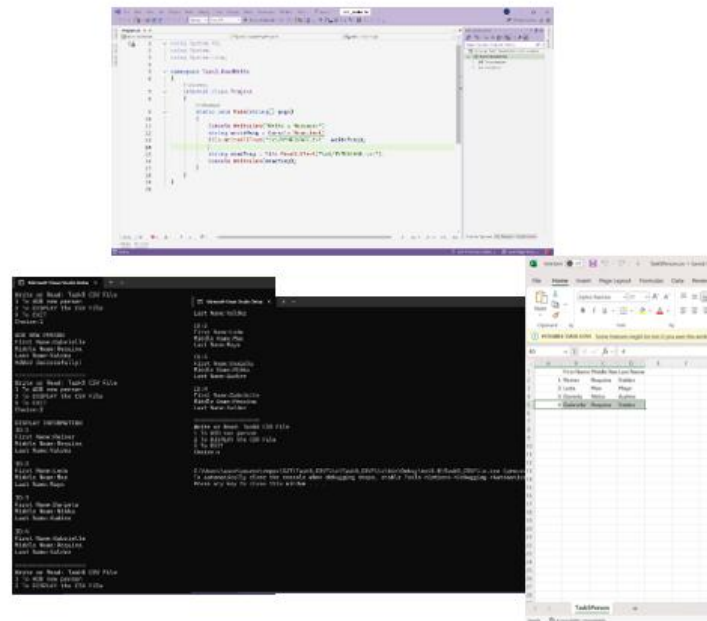


Figure 4. Text and CSV file

This system is the creation of a .txt and CSV file by entering an input in a console app (See Figure 4). The C# console will ask the user to input a message to be stored in a text file, creating a message and first, middle, and last name will be inputted to be stored in a spreadsheet, CSV file. In addition, a simple navigation was put in with a display option to show the data that was stored in CSV. JSON was installed in Visual Studio Community for this process.

DATABASE MANAGEMENT USING MONGODB



Figure 5. CRUD with MongoDB

Figure 5 presents creating a CRUD application for the data using MongoDB and displaying it on a console app. Menu-like options were used for easy navigation. Each number is equal to Create, Display, Update, and Delete, with the additional number for exit of the console. The data that was used came from the previous Task that was stored in a CSV spreadsheet.

CREATION AND TESTING OF APIs

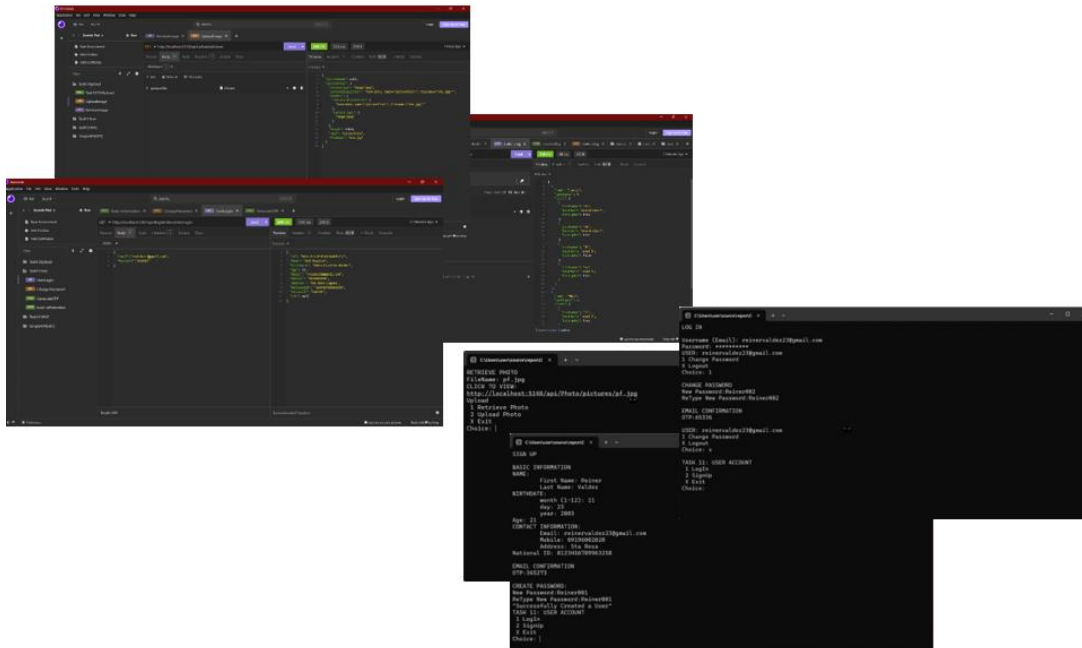


Figure 6. Testing of APIs using Insomnia and Console App

Figure 6 illustrates the use of Insomnia and console for testing APIs on the MVC. Familiarizing with the framework, MVC, and creating simple to complex API endpoints. The APIs were called and tested on Insomnia to ensure that the connection to the APIs was running properly. Post, Put, Delete, and Get were used for testing the sample API's endpoints.

One of the sample tests is the Uploading of a CSV file and a picture. The pictures were uploaded locally and not to the database.

Combining the User Registration, a previous task, and uploading of file solution, the APIs were called in the console app and task to use a RESTful API. To navigate on both system a simple navigation were used in this system leading to either User Registration or uploading of file.

REAL-TIME CHATHUB USING SIGNAL R (CONSOLE, WEB, MOBILE, WINFORMS)

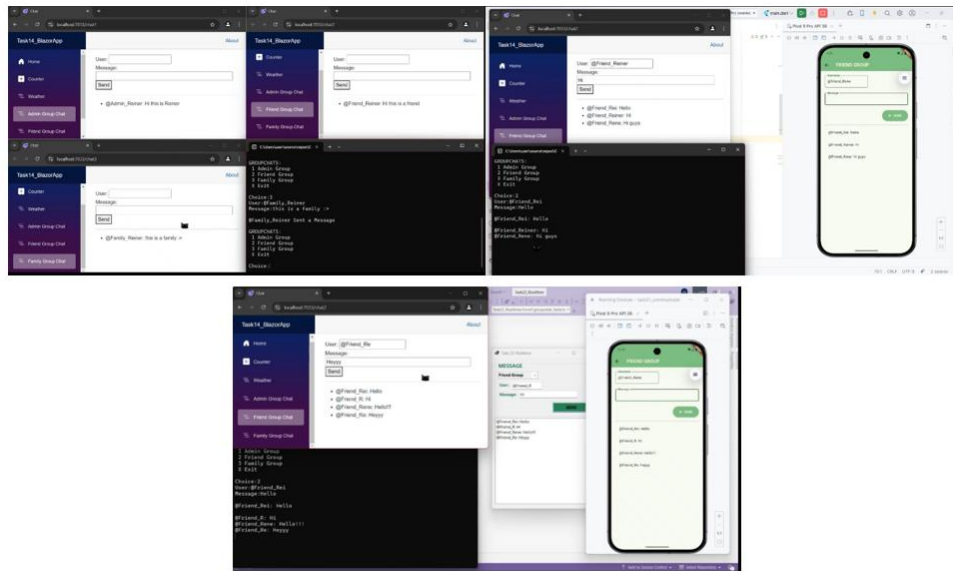


Figure 7. ChatHub using Web, Console, Mobile, Windows Form

Introducing the Signal R, a library adding real-time Functionality to Web applications. Blazor app was used to create the system of web applications. We created 3 group chats named Admin, Family, and Friends, which will be able to communicate with each other in real time, as shown in Figure 7.

With SignalR, creating proper communication while still having 3 groups and creating a connection between various platforms on Web, Console, Windows Form, and Mobile.

DEVELOP A SIMPLE MOBILE APP USING FLUTTER

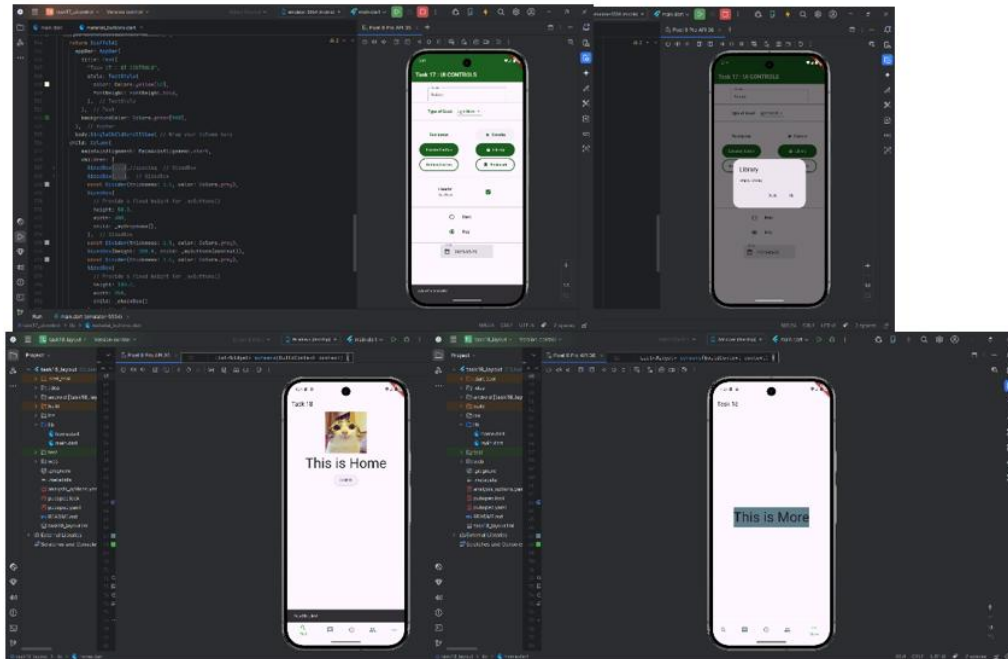


Figure 8. User Interface using Flutter in Android Studio

Using software tool Android Studio, or use a visual studio, they created simple User Interface to familiarize in syntax, controls and such. On the one mobile app, in one page various controls were put together like, button, text field, dropdown, dateTime and more.

Continuing the familiarization in Flutter, another mobile app was created to have a bottom navigation with each page having a simple text, and one of them has an image that has a button, and a snack bar will appear at the bottom.

USER REGISTRATION (CONSOLE, WEB, MOBILE)

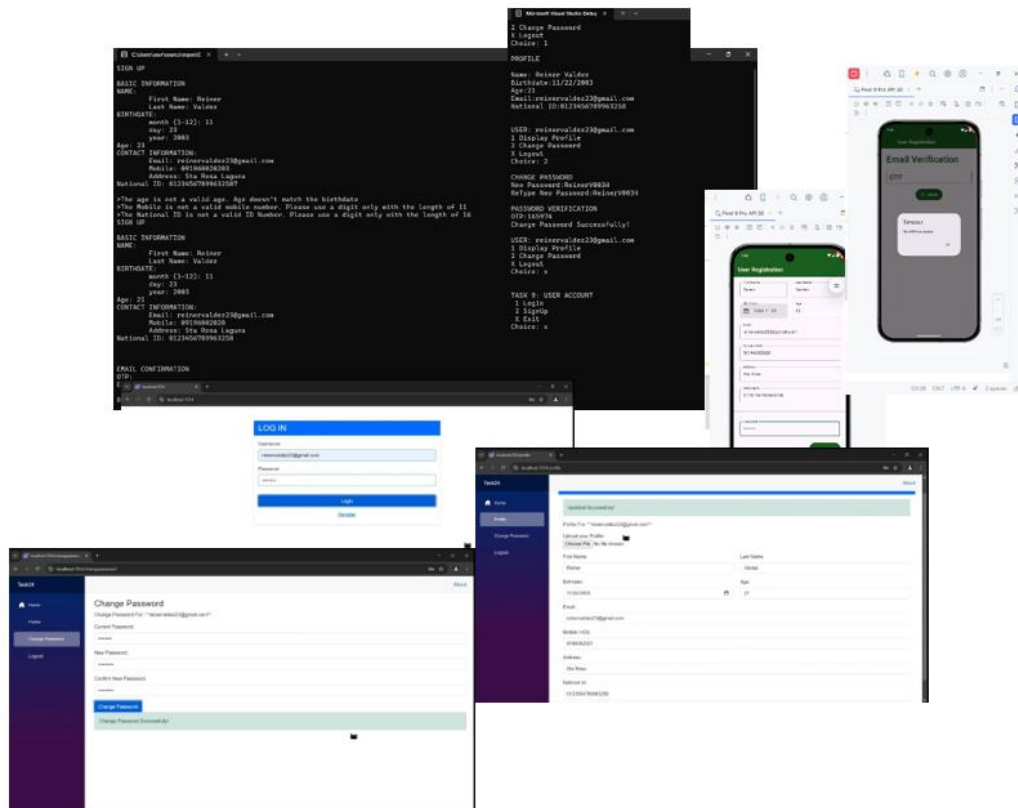


Figure 9. User Registration in console, web, mobile

The User Registration system was created on various platforms together with a database of MongoDB (see Figure 9.). There is an OTP in change password and registration, it was sent through email that has an expiration of 2 minutes. In Console app, the system can resend the OTP once the time expired. There is also a Login option displaying simple information and where the change password resides.

Following the C# console app the mobile user registration was created moreover a backend was created using MVC, that was published. These Mobile app only has user registration sending OTP through Email.

Lastly, Blazor app for web application similar function with an addition of display and update profile. A JWT Token was used to ensure the Authentication and Authorization of the User.

AUTHENTICATION AND AUTHORIZATION USING JWT TOKEN

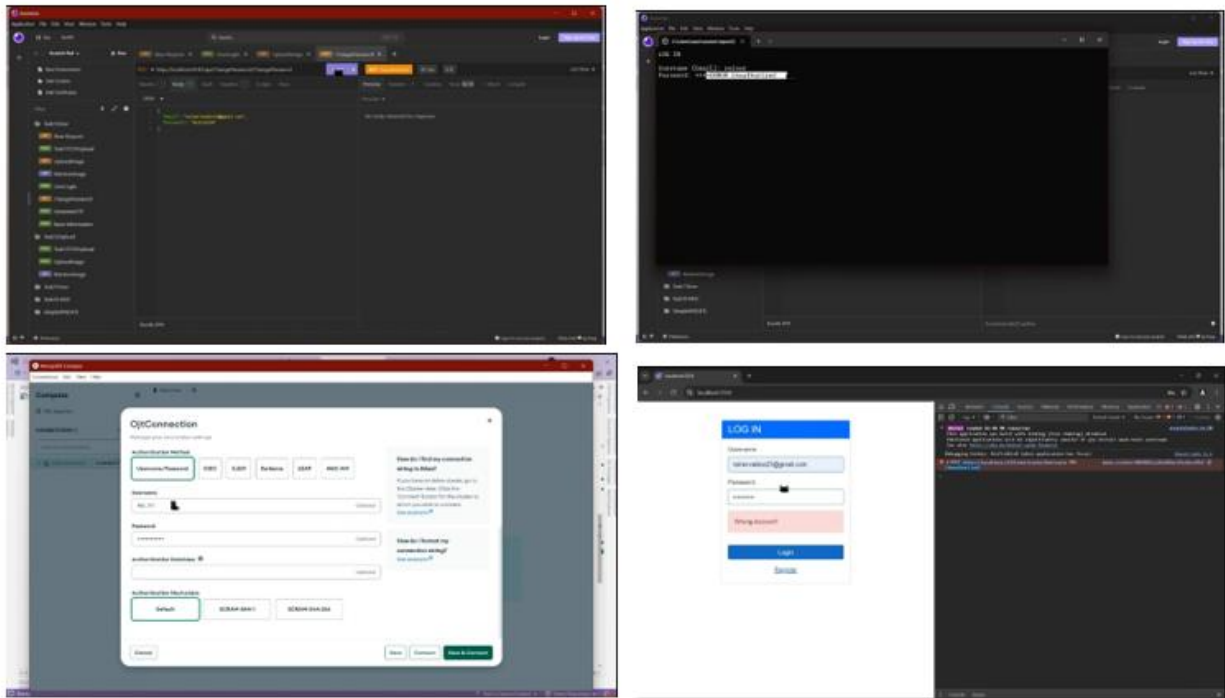


Figure 10. JWT Token testing and Authenticate Database

This task was focused on Authorization and Authentication using JSON Web Token and adding a username and password to the Database. Shown in Figure 10, a 401 Unauthorized response will appear if the user doesn't have the right credentials and tries to access pages before logging in. The authorization was tested on Insomnia, Console, and web.

MODULE 3 CHAT MESSAGES

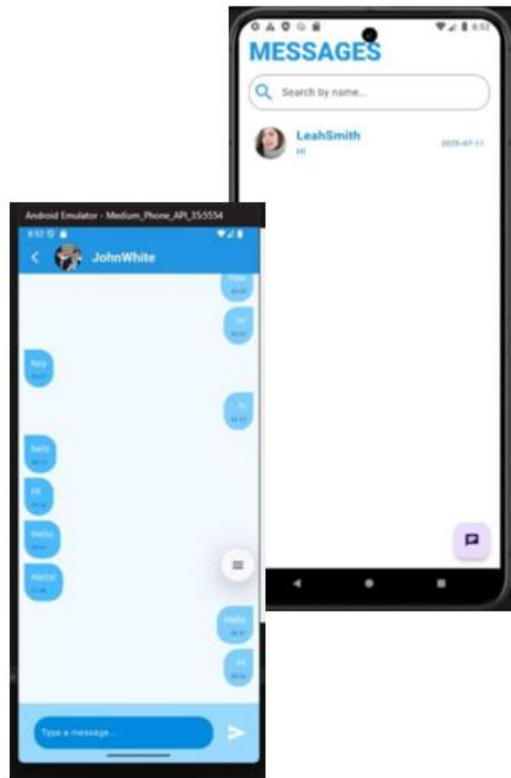


Figure 11. Initial Module 3 Messages

Figure 11 represents the initial development of the mobile app for the message or chat module, creating a real-time message between users, Parker, and the owner. This has 2 pages, a chat list, and a message page. Chat list is the collection of conversations of the users, it has the functionality of search by name, notification, or highlighting when a new message is sent. The message page consists of 2 bubbles between the users. The User Interface was created with the idea of a simple and clean look while adapting to the color of the iNeed Solution Inc. For testing of this module, a temporary user was used.

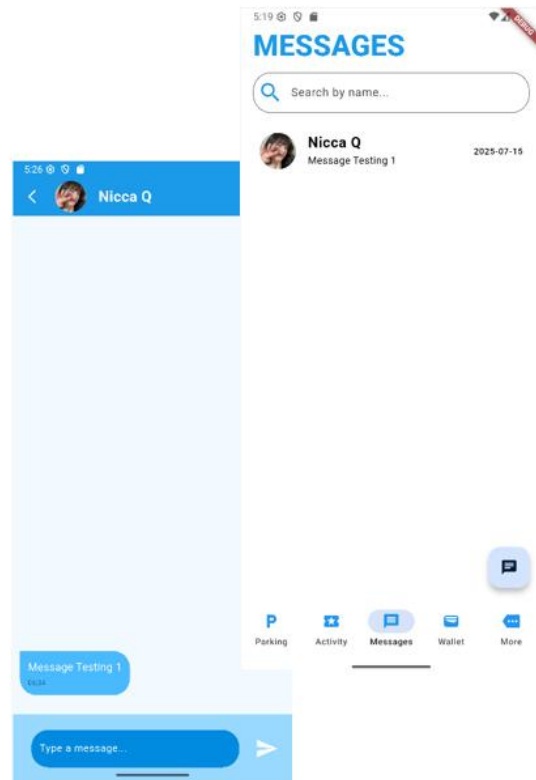


Figure 12. Current Module 3 Message

This is the current look of the mobile after the merging of the modules. The software tools that was used for merging is GitHub. The temporary hardcoded user for testing were removed and exchanged with the token of the user once logged in.

SYNTHESIS OF THE PRACTICUM

LEARNING GAINS

While undertaking her practicum, Reiner gained significant knowledge, encompassing both technical and soft skills. For technical skill, she was able to effectively use various software tools, frameworks, and others that she wasn't familiar with, had never used before, or rarely used. These include:

- MongoDB: Learned to implement and manage a Database with MongoDB
- SignalR: Acquired skills in the application of a real-time functionality
- MVC: Developed a whole architectural application.
- BlazorApp: Built a Web Application using Microsoft's Blazor Framework
- Insomnia: Gained skill for API Testing and Development
- Draw.io: Applied for creating various diagrams and flowcharts for the collaborative Project design.
- Android Studio: Utilized in developing and debugging Android applications.
- Flutter & Dart: Gained Proficiency in building a mobile application
- GitHub: Improved collaborative coding practices
- AnyDesk: Gained Experience in remote desktop access and control, enhancing my ability to manage from a distance.
- Task-Scheduler

Beyond the Technical skills, she significantly improved in soft skills. She learned to collaborate and communicate better with my peers during team projects and gained better confidence in asking and socializing more. This practicum also refined her organizational skills and gained better coding practices. These Learning gains have broaden her knowledge and experience that could contribute for her future professional development.

REALIZATION

During the duration of working in the Company of iNeed. She experienced and learned various tools and frameworks that in relatively short time she was able to have acquired many technical

skill besides this even though it was a work from home with a limited of interaction with the other trainees and employees she were able to collaborate more gaining confidence in her social skills such as she become more communicative and asking inquiries from her superiors and other trainees. Her organization in files and codebase upskilled; moreover, with a fast-paced environment, she was able to adapt more than she expected, despite being busy with other work. Because of the training plan of iNeed, it reminded her again that the simpler the task, the harder it was. She overthinks those simpler tasks as she got used to complex tasks, as those simpler tasks were the foundations for more challenging tasks, she thought that she needed to hone and work them more.

CONCLUSION

In conclusion, this practicum at iNeed proved to be an invaluable experience, significantly expanding both her technical proficiency and essential soft skills. It was a journey from basic tasks to more complex challenges, which not only solidified her existing knowledge but also cultivated new knowledge. This experience instilled greater confidence and a more strategic approach to problem-solving, preparing her for future professional endeavors in software development.

APPENDICES

Appendix A

Competency-Based CV

Reiner Gabrielle R. Valdez

reinerrvaldez@live.mcl.edu.ph | (+63)919-600-7012
Sta Rosa City, Laguna.

EDUCATION

Mapúa Malayan Colleges Laguna

2021 - Present

B.S. Information Technology

Cabuyao, Laguna

- Consistent Dean's Lister
- **Relevant Coursework:** Web & Mobile App Development, Database Management, Embedded System, IoT.

Mapúa Malayan Colleges Laguna

2019 - 2021

Senior High - Information Communication Technology

Cabuyao, Laguna

- Graduated with Principal's List for Academic Excellence Award
- **Relevant Coursework:** Mobile Android Development, Adobe Photoshop, Animate, Illustration

EXPERIENCE

iNeed Solutions Inc.

April - July 2025

Junior Software Engineer Intern

Makati City, Philippines

- Intern at iNeed Solutions Inc., completing approximately 486 hours.
- Focused on developing, debugging, designing, and writing clean, maintainable code.

PROJECTS

iNeed Parking Application

June - July 2025

Junior Software Engineer Intern

- Built the front and back end of the messaging section in the mobile application using Android Studio (Flutter), Visual Studio Code (MVC, SignalR) and MongoDB.
- A Parking Management application that involves both mobile and web for administrative and user-side. Allows users to view and locate parking areas, manage session, process payment, activity logs and communicate through the built in-chat function.

Air Quality Monitoring System

January - July 2025

Member - Web Developer

- Designed a responsive website to display real-time logging system for tracking and analysis using Visual Studio Code (HTML, CSS, JavaScript) and PHP Script.
- An IoT-based Air Quality Monitoring System using MQ Sensors and ESP32 to detect gases together with Arduino IDE for programming.

Smart access control and Monitoring System

June 2024 - July 2024

Member - Mobile App Developer

- Developed a Mobile Application to assist in controlling the Lights and sensors, additionally it displays and store all data entries with Xamarin Android Studio in Visual Studio 2022(XML, C#) and PHP Script.
- Implements a system in an establishment that can be accessed only by authenticated and authorized people using RFID sensor, LED and Buzzer.

VOLUNTEERING

CALABARZON: IT Cybersecurity Caravan - Equipment Committee

June 2024

- Voluntary assistance on technical parts of the event. Assisted with the management of online call in Zoom and Manage Speaker's document.

CERTIFICATE, SKILLS & INTERESTS

- **Certificate:** CompTIA IT Fundamentals (ITF+) Certification
- **Languages:** HTML, CSS, JavaScript, C#, PHP, Java, XML (Familiar), Python (Familiar), C (Familiar)
- **Framework/Libraries:** Asp.net, Leaflet, Chart.js, Swiper
- **Database Management:** MySQL, MongoDB, SQL Server (Familiar)
- **Software Tools:** Visual Studio 2022, Visual Studio Code, Android Studio, Figma
- **Soft Skills:** Strong Problem-Solving, Effective Communication, Team Collaboration, Adaptability, Time Management, Learning Dedication
- **Interests:** Reading; Drawing/Sketching; Movies/Series; Playing Casual, Mystery or Puzzle Games.

Appendix B

Endorsement Letter



5 May 2025

Dhalia Cardines

Corporate IT and Administration

iNeed Solutions Inc

2nd Floor, Wyce Business Center 9599 Kamagong Barangay San Antonio

Makati City 1203, Philippines

Dear **Ms. Cardines**,

The B.S. in Information Technology program of Mapúa Malayan Colleges Laguna requires their students to undergo Practicum program for a minimum of **486** hours in an academic calendar that will prepare our students to be job-ready after completing their curriculum. This program intends to enable our students to acquire and practice the knowledge and skills expected of a graduate of a B.S. IT program which, in turn, would guarantee continuous supply of IT professionals needed by your company.

We believe that your company can provide the relevant exposure necessary for our students to achieve the intended learning outcomes for the B.S. in Information Technology program. In this regard, I would like to endorse **Mr. Reiner Gabrielle R. Valdez** to have his practicum activities in your company as requested.

We thank you for your confidence and trust with us and we look forward to a more meaningful linkage that is mutually beneficial to our students and your company.

With warm regards,

A handwritten signature in black ink, appearing to read 'Adomar L. Ilao'.

ADOMAR L. ILAO, DIT

BSIT Program Chair

College of Computer and Information Science

Mapúa Malayan Colleges Laguna

alilao@mcl.edu.ph

(049) 832-4076

Appendix C

Practicum Confirmation and Acceptance Form



REVISION NO.: 00
REVISION DATE: May 10, 2016

PRACTICUM CONFIRMATION AND ACCEPTANCE FORM

IMPORTANT INFORMATION

- STUDENTS ACCEPTED FOR PRACTICUM IN A HOST COMPANY WILL HAVE TO ACCOMPLISH THIS FORM.
- ASK THE PRACTICUM SUPERVISOR/ COMPANY REPRESENTATIVE TO FILL IN THE DETAILS OF THE TRAINING.
- SUBMIT TO THE PRACTICUM ADVISER/COORDINATOR PRIOR TO THE START OF TRAINING.

NAME OF STUDENT	Reiner Garielle R. Valdez	STUDENT NUMBER	2021160610
COURSE CODE	IT199F	SYTERM ENROLLED	2024-2025/3T

This is to certify that Reiner Gabrielle R. Valdez (name of student-trainee) has been accepted for practicum at Reed Solution Inc., 2nd Floor, Wyce Business Center 9599 Kamagong Street, San Antonio Makati City 1203, Philippines (name and address of establishment) and will be attached to the Information Technology department/s for a minimum of, but not limited to 486 hours. Training will commence on April 22, 2025 and is expected to end on July 22, 2025. Attached is the list of requirements.

COMPANY REPRESENTATIVE	
<u>Dhalia T. Cardines</u>	<u>Admin</u>
Signature over Printed Name	Official Designation
<u>Engineering Department</u>	<u>admin@ineed.com.ph / 0917-771-7275</u>
Department	Email and Contact Number/s

NOTED BY	<u>5/2/2025</u>
<u>Adrian Diao</u>	Date
Signature over printed name of Practicum Coordinator	

COPY: (1) STUDENT, (2) HOST COMPANY, (3) PRACTICUM COORDINATOR

FORM OVPA 030B

THIS FORM IS AVAILABLE AT THE OVPA.

Appendix D

Liability Waiver



Malayan Colleges Laguna
A MAPUA SCHOOL

REVISION NO.: 00
REVISION DATE: May 10, 2016

STUDENT TRAINING AGREEMENT AND LIABILITY WAIVER

IMPORTANT INFORMATION

- THIS FORM IS TO BE ACCOMPLISHED AND SUBMITTED BY STUDENT TRAINEE TO THE PRACTICUM ADVISER BEFORE STARTING THE PRACTICUM.
- READ AND UNDERSTAND THE PROVISIONS OF THIS AGREEMENT AND WAIVER.
- ENSURE THAT ALL SIGNATORIES SIGN THE FORM.

I, Reiner Gabrielle R. Valdez, and a student of MALAYAN COLLEGES LAGUNA (hereinafter referred to as "MCL", do hereby voluntarily undergo on-the-job training at iNeed Solution Inc., hereinafter referred to as the "Host Company", located at San Antonio Makati City 1203, Philippines, under the following terms and conditions:

- That the practicum training will commence on April 22, 2025 and ends on July 22, 2025 and will have to complete a minimum of 486 hours required for the on-the-job training;
- That I shall observe proper decorum and act professionally at all times and abide by the Company's rules and regulations and comply with those imposed for the training program, otherwise, I shall be excluded from further participation;
- That in the course of my training program, I may have access to information which may be of confidential in nature and proprietary to the Company, for which I may be required to execute a confidentiality and non-disclosure agreement as a prerequisite to my participation in the training program;
- That the time I will spend on the training program in the completion of my on-the-job training requirements will not and should not be interpreted or construed as working hours and should be regarded as non-compensable. Provided that, the Company may, as a unilateral act of liberality or generosity on their part, provide me with meal, travel, transportation allowances, accommodations, etc.;
- That I fully understand that notwithstanding the allowances enumerated in the preceding section which I may receive, there exists no labor-management and/or employer/employee relationship between me and the Company where I will undergo my training;
- That I shall exercise due care and diligence in the tasks assigned to me and personally be made answerable for any and all liabilities for damage to property or injury to third person, which may be occasioned by my intentional or negligent acts during the course of my on-the-job training;
- That I shall likewise hold the Host Company and MCL free and harmless from any and all liability and responsibility for any sickness or injury to myself and third parties and damage to property which I may sustain and/or may occur at any time during the training program, including time spent in traveling to and from any and all premises and locations where I may be required to go to as part of my training program;
- That the Company reserves the right to discontinue my training on reasonable grounds upon written notice to MCL and myself. Additionally, in the event my training program is discontinued for reasons attributable only to myself, I may be made to reimburse the Host Company for any/all the allowances, stipends, etc., which I may have received from them during and prior to the termination of my training program;
- That in addition to my liability under section *g* and for the pre-termination of my training program provided for under section *h* hereof, I may be subjected further to disciplinary action in accordance with the school's student manual and/or be a ground for disqualification from graduation;

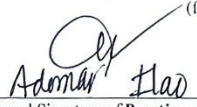
Signed on this 23 day of April 2025.


Reiner Gabrielle R. Valdez
Signature over printed name of Student Trainee

WITH OUR CONSENT:

Signature over printed name of Parent/Guardian
(for minors only)

NOTED BY:


Printed Name and Signature of Practicum Adviser/ Coordinator


Printed Name and Signature of Host Company Representative

COPY: (1) STUDENT; (2) HOST COMPANY; (3) PRACTICUM ADVISER; (4) PRACTICUM COORDINATOR

FORM OVPA 030C

THIS FORM IS AVAILABLE AT THE OVPA.

Appendix E

Training Plan



REVISION NO: 00
REVISION DATE: May 10, 2016

TRAINING PLAN

NAME	Reiner Gabrielle R. Valdez	COURSE CODE	IT199F
PROGRAM & STUDENT NO.	BSIT - 2021160610	COURSE TITLE	IT PRACTICUM

STUDENT OUTCOMES CO1. Identify, analyze, and design business process solutions to the problem faced by the organization. CO2. Apply the different concepts of systems analysis and design, software engineering, database management, and programming courses in the problem-solving process in the organization, and CO3. Acquire new knowledge and experience while in the organization.
--

AREAS / PHASES OF TRAINING AND TIME ALLOTMENT	
A. Company Orientation / Training Orientation B. Software Development (but not limited to Research and Implementation) C. Career Planning and Skills Development D. Technical Documentation	- 8 hours - 382 hours - 80 hours - 16 hours

EVALUATION GUIDELINES & COURSE OUTCOMES	
DEMONSTRATION OF SOFT SKILLS (40%) KEY AREAS COMMUNICATION SKILLS (20%) Relate to co-trainees/supervisors terminologies and rules Recite procedures and instructions needed for the tasks Identify and describe safety signs and symbols Ask critical questions related to the tasks Produce well-written regular and incident reports Prepares and presents reports using Information and Communication Technology (ICT) PROFESSIONAL DEPORTMENT (20%) Observes proper grooming and attire Reports to work regularly on time and as necessary, even beyond prescribed working hour Acts according to the job description given by the company Willing to accept new tasks apart from the usual routine and responsibilities Delivers quality output on time Demonstrates respect for different individuals INITIATIVE (+5%) Volunteers to perform tasks beyond routine tasks	DEMONSTRATION OF TECHNICAL SKILLS (60%) KEY AREAS Demonstration of Technical Skills (40%) • Able to demonstrate basic programming skills in problem-solving and complete short tasks, with knowledge of fundamental paradigms (10%) • Able to use database tools and techniques to accomplish given data management tasks efficiently and accurately. (5%) • Able to identify, analyze, and resolve issues or bugs during development. (5%) • Able to build mobile applications, interact with APIs, and create user interfaces for both desktop and console applications (20%) Documentation and Skill Development (20%) • Able to write documentation of various technical aspects, including functionality, architecture, design, and usage with clarity and consistently (5%) • Able to understand and apply foundational Project Management principles practically, integrating and adapting to emerging technologies in professional settings. (10%) • Able to set career goals, understand personal strengths, explore career paths, engage in self-discovery and develop leadership and management skills (5%). INITIATIVE (+5%) Volunteers to perform tasks beyond routine tasks

CONFORME	CONSENT (FOR MINORS ONLY)	NOTED BY	ENDORSED BY	APPROVED BY
 Reiner Gabrielle R. Valdez / 04-26-2025 SIGNATURE OVER PRINTED NAME OF STUDENT / DATE	 SIGNATURE OVER PRINTED NAME OF PARENT OR GUARDIAN / DATE	 Jonarc S. Cardines / 29Apr25 SIGNATURE OVER PRINTED NAME OF PRACTICUM SUPERVISOR / DATE	 7/17/2025 SIGNATURE OVER PRINTED NAME OF PRACTICUM ADVISER / DATE	 7/17/2025 SIGNATURE OVER PRINTED NAME OF PROGRAM CHAIR / DATE

COPY: (1) STUDENT; (2) HOST COMPANY; (3) PRACTICUM COORDINATOR

FORM OVPAA-030D

THIS FORM IS AVAILABLE AT THE OVPAA.

Appendix F

Weekly Journal



REVISION NO.: 00
REVISION DATE: May 10, 2016

DAILY JOURNAL

IMPORTANT INFORMATION

- INCLUDE TASK ASSIGNMENTS OR MOVEMENTS, REFLECTION ON THE DAY'S NEW LEARNING, ACCOMPLISHMENT, CHALLENGES FACED AND HOW YOU RESPONDED, OBSERVATIONS AND RECOMMENDATIONS ON THE IMPROVEMENT OF SYSTEMS / OPERATION / MANAGEMENT, ETC.
- SCANNED COPIES OF THIS FORM SHALL BE SUBMITTED ON A WEEKLY BASIS THROUGH APPROVED LMS.
- HARD COPIES OF THIS FORM SHOULD BE COMPILED AS PART OF THE STUDENT'S PORTFOLIO.

DATE	Week #1: April 22 – 25, 2025	AREA ASSIGNMENT	Console Application Development
TASK	Task 1 - 4	SHIFT/TIME	8:00 AM – 5:00 PM

TRAINING ORIENTATION:

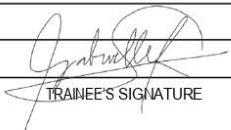
Earlier in the first week, we were oriented on the regulations to follow and how to document our OJT schedules. This includes the Schedule Planning and the time in and out, as well as the workday summary, and the meetings on Tuesday and Friday. Additionally, we had a brief conversation about introducing the trainees and other employees.

SOFTWARE DEVELOPMENT:

We were tasked with 4 Individual tasks. Task 1 is simple reading in W3School and doing the exercises in them on the topic of C#, together with this is following the document named 'Pre-on Boarding', which involves downloading, creating accounts, and software tools that will be used for the whole training. Task 2 is a C# console application, create 5 patterns that will ask the user to input a character and how many of them there are. Task 3 is regarding a read/write exercise, and Task 4 is to create a Task Scheduler that writes to a file every 1 hour, using Visual Studio as its .exe or program.

TECHNICAL DOCUMENTATION AND SKILLS DEVELOPMENT:

This week, the Liability Waiver and Acceptance Form was sent to them via email to be signed, and they were able to receive it that week. A brief meeting was held to give us a heads-up regarding the tasks and discussions. We had simple exercises using the C# Console Application, reminding of it, also this introduced me in creating a Task in Task Scheduler.



TRAINEE'S SIGNATURE

DAILY JOURNAL

IMPORTANT INFORMATION

- INCLUDE TASK ASSIGNMENTS OR MOVEMENTS, REFLECTION ON THE DAY'S NEW LEARNING, ACCOMPLISHMENT, CHALLENGES FACED AND HOW YOU RESPONDED, OBSERVATIONS AND RECOMMENDATIONS ON THE IMPROVEMENT OF SYSTEMS / OPERATION / MANAGEMENT, ETC.
- SCANNED COPIES OF THIS FORM SHALL BE SUBMITTED ON A WEEKLY BASIS THROUGH APPROVED LMS.
- HARD COPIES OF THIS FORM SHOULD BE COMPILED AS PART OF THE STUDENT'S PORTFOLIO.

DATE	Week #2: April 28 – May 2, 2025	AREA ASSIGNMENT	Console Application Development
TASK	Task 5 - 9	SHIFT/TIME	8:00 AM – 5:00 PM

SOFTWARE DEVELOPMENT:

For the second week, various tasks were given, focusing on console apps and the database. For the Task 5 we were given a task to create a csv file with C#, Task 6 is an automated task scheduler Email that will happen every hour for the whole day, the visual studio console application were used for this program, Task 7 use of interface class using the previous task, Task 8 is creating a CRUD with Mongo DB using the topic or model person and lastly Task 9 a User Registration, that is connected to database, has an OTP that were sent through Email for Change Password and Email Verification, the OTP also expired after 2 minutes. For the expiration, it is solved by putting a timeout on the Console.ReadLine.

TECHNICAL DOCUMENTATION:

The Training Plan was sent this week and also received this week, signed. A meeting on Tuesday and Friday occurred, giving us a new Task and giving a short lesson for the next task; moreover, it includes the topic of the situations of our current task. I had difficulty in creating a Timeout for OTP but eventually solved it in this task is helps be more organized in coding assisting me in enhancing my code practices.



TRAINEE'S SIGNATURE

DAILY JOURNAL

IMPORTANT INFORMATION

- INCLUDE TASK ASSIGNMENTS OR MOVEMENTS, REFLECTION ON THE DAY'S NEW LEARNING, ACCOMPLISHMENT, CHALLENGES FACED AND HOW YOU RESPONDED, OBSERVATIONS AND RECOMMENDATIONS ON THE IMPROVEMENT OF SYSTEMS / OPERATION / MANAGEMENT, ETC.
- SCANNED COPIES OF THIS FORM SHALL BE SUBMITTED ON A WEEKLY BASIS THROUGH APPROVED LMS.
- HARD COPIES OF THIS FORM SHOULD BE COMPILED AS PART OF THE STUDENT'S PORTFOLIO.

DATE	Week #3: May 5 – 9 ,2025	AREA ASSIGNMENT	Backend Development
TASK	Task 10 - 13	SHIFT/TIME	8:00 AM – 5:00 PM

SOFTWARE DEVELOPMENT:

For the third Week, the focus is on using MVC on Visual Studio 2022 and using it together with the Console application; the past Tasks were used to create some of it, like user registration. Starting on Task 10, a simple task, and endpoints were created using MVC, to familiarize with the framework more, and usage of Insomnia, also to create Endpoints for the MVC and call them from the console app. Task 11 was to create an MVC user registration, creating the simple endpoints in MVC and using Insomnia to check the API. Task 12, to be able to upload files like images and CSV, and also to be able to retrieve them on the console app; furthermore, endpoints were called from the console app to the MVC. Task 13 is the restructuring of the API using Tasks 11 and 12, combining them and creating a call to endpoints from the console app to MVC. A requirement to use a RESTful API was added, and a simple navigation was added, such as adding menu options and a back option for easier navigation.

TECHNICAL DOCUMENTATION AND SKILL DEVELOPMENT:

At the meeting, we were briefed and taught how to manage an endpoint, which includes creating, using, and connecting them to another application, and creating a well-developed back-end. We were also introduced to another tester application called Insomnia making it easier for checking the endpoints. With the restructuring of task this help me to have a better coding practices and creating a cleaner coding.



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DATE	Week #4: May 13 – 16 ,2025	AREA ASSIGNMENT	Web and Mobile Application Development
TASK	Task 14 - 18	SHIFT/TIME	8:00 AM – 5:00 PM

SOFTWARE DEVELOPMENT:

For the fourth week, we were introduced to Signal R and were given a task, creating a group chat and being able to pick a key on the username. We need to connect it to the console app again, which was for Task 14. Task 15 is the introduction to Flutter, and Task 16 is the design for UI Wireframe, making us create a flow diagram and wireframe for it. The design is about the parking owners and parking slots. For Task 17, since we were introduced to Flutter, we needed to be familiarized with the syntax, controls, and such, so we were given a task to create a page with various UI controls. Task 18 is from task 16 with the same flow. We need to create the bottom navigation and put a layout on one of them, and put a simple text on it.

TECHNICAL DOCUMENTATION AND SKILL DEVELOPMENT:

In regard to newly Introduced Library in ASP.NET Core, SignalR. In the meeting, we were shown the demo of how SignalR is being used, and with that, we applied it to the newly assigned task. In building a cross-platform application, we follow the installation for Flutter and Dart as we will be using it as a framework for developing a mobile application. This framework really broaden my knowledge bringing in that would help a real-time system application.


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DATE	Week #5: May 19 – 23 ,2025	AREA ASSIGNMENT	Web and Mobile Application Development
TASK	Task 19 - 22	SHIFT/TIME	8:00 AM – 5:00 PM

SOFTWARE DEVELOPMENT:

For this week, we focused on Flutter. We were tasked to download it, and I used Android Studio instead of Visual Studio Code. For Task 19, it uses IIS for Task 13; besides this, the others are for Flutter. Task 20 is User Registration using the Task 13 MVC endpoints, connecting it to the database. Task 21 Group Chat connecting the other like console app, Blazor App, and Task 22, the task 21 with WinForms.

TECHNICAL DOCUMENTATION AND SKILLS DEVELOPMENT:

The development of mobile can be done on Android Studio or Visual Studio Code, and I used Android Studio. Because of the use of IIS, we were able to use the backend without needing to run it. This week, we utilized SignalR, connecting it on various platforms, introducing me to hubconnection or Hubs, and assisting me in broadening my understanding of it. Moreover, the usage of Flutter it help in providing a seamless user experience and with it it was easy to use or apply despite only using it for a while.



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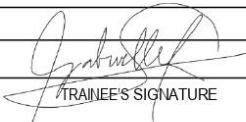
DATE	Week #6: May 26 – 30 ,2025	AREA ASSIGNMENT	Web Application Development
TASK	Task 23 - 24	SHIFT/TIME	8:00 AM – 5:00 PM

SOFTWARE DEVELOPMENT:

For this week, we focused on authentication using JWT and MongoDB and creating a user registration in a Blazor app while still using the Task 13 MVC as a backend API, adding the Task 23 JWT too for the authentication. We focused more on the authorization and Authentication for these tasks. For the Authentication and Authorization, '401' should appear when trying to log in. The generated JWT was used for this and authorized in the M. It tried to use Insomnia, console, and web application as mentioned. The Database also has a username and password.

TECHNICAL DOCUMENTATION AND SKILLS DEVELOPMENT:

This week, I learned more about authorization and Authentication, as well as a further understanding of status codes. Not only on web applications, but also the authentication and authorization in the Database by putting a username and password on it.



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DATE	Week #7: June 2 – 5 ,2025	AREA ASSIGNMENT	Web application development
TASK	Task 23- 24	SHIFT/TIME	8:00 AM – 5:00 PM

SOFTWARE DEVELOPMENT:

This week, we are finishing all the tasks that were given to us from 1 – 24 to make sure everyone catches up. We also discussed the groupings for the new task, and we discussed the modules and their distributions. The materials for the task were shared, and we were asked to clone the repository on our local computer. Furthermore, this week I submitted the 23rd and finished Task 24. Task 24 is a user registration in a web application using the Blazor app while communicating with task 13 MVC backend API. Also, using JWT as a token to log in and be authenticated. It has the feature of the past Task User Registration Module, with the addition of Update and viewing of the Profile.

TECHNICAL DOCUMENTATION AND SKILLS DEVELOPMENT:

For Task 24, I had the most challenging as JWT, authorization, and Blazor App were areas I am not yet familiar with. However, I was able to finish it after a lot of trial and error. This difficulty enhances my knowledge and perseverance in challenging these difficult activities.



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DATE	Week #8: June 9 – 13 ,2025	AREA ASSIGNMENT	Software Development
TASK	Planning	SHIFT/TIME	8:00 AM – 5:00 PM

SOFTWARE DEVELOPMENT:

For this week, we are starting the planning and finalizing the assigned task we were given. We were assigned to create User stories, use case diagrams, UI frameworks, ERD, class diagrams, flow charts, and a List of API. It was reviewed by the supervisor, and we were asked to redo some parts. Research and in-depth understanding were done to create the finalized plan.

TECHNICAL DOCUMENTATION AND SKILLS DEVELOPMENT:

Because of this planning, it reminded me of the clear importance of planning, especially in programming. Also, it enhances my skills in diagrams, use cases, and overall planning of the module.



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DATE	Week #9: June 16 – 20 ,2025	AREA ASSIGNMENT	Software Development
TASK	Proof of Concept	SHIFT/TIME	8:00 AM – 5:00 PM

SOFTWARE DEVELOPMENT:

Following Last Week Planning of the Project, that we were assigned to. I was asked to create a POC or Proof of concept to see the possibility of the planned put in the Draw.io. The POC contains of developing a working backend and using the console as a temporary User Interface to display a proper working backend also tested on Insomnia. I used the localhost MongoDB as my temporary database for this task. The past task was a helpful guide as we were using framework and software tools.

TECHNICAL DOCUMENTATION AND SKILLS DEVELOPMENT:

I learned about Proof of Concept, this is to create a experimental project or build a demonstration to view the feasibility of our created Plan. Similar to our individual Task. The way on how to submit, we need to send the Code, Output Screenshot and Video explaining the code or how it works. Because of the created plan it was easier for me to create a POC and pass it more in time.



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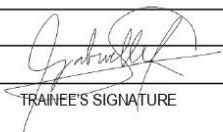
DATE	Week #10: June 23 – 27 ,2025	AREA ASSIGNMENT	Software Development
TASK	Modification and GUI	SHIFT/TIME	8:00 AM – 5:00 PM

SOFTWARE DEVELOPMENT:

For our assigned portion of the project, I modified the feedback regarding the coding convention of our supervisor in the Teams such as the naming and the structure of the coding. I tasked my member to create the initial component of the mobile User interface following the UI/UX I designed for the mobile. I also shared the refined backend code via Google Drive and provided other additional sample code relevant for our project.

TECHNICAL DOCUMENTATION AND SKILLS DEVELOPMENT:

Earlier this week, I submitted the Proof of Concept (POC) and gained the feedback of our supervisor. He raised few inquiries concerning the video or code which I address immediately thus allowing our team to proceed in creation of the main project. I gain better confidence in communicating with others to collaborate with them.



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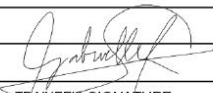
DATE	Week #11: June 30 – July 4 ,2025	AREA ASSIGNMENT	Software Development
TASK	Project Development	SHIFT/TIME	8:00 AM – 5:00 PM

SOFTWARE DEVELOPMENT:

Continuing from last week our team successfully implemented the core functionality of the project and its properly working with minimal issues or problems. With the main function working integrated with the backend and its localhost MongoDB, we were able to shift focus on the more detailed section of our assigned part of the Project. We refined the designed to make sure it follows the appearance of the UI/UX that was in the plan. Identifying the issues in the codebase we updated and modified the backend to make sure its functioning smoothly.

TECHNICAL DOCUMENTATION AND SKILLS DEVELOPMENT:

Our Team actively collaborated and maintained consistent communication, facilitating smooth sharing of the development of the project and the modified works. This ensured that both of us were on the same track and page to avoid any problems also creating efficient working system for both of us.



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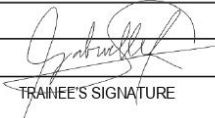
DATE	Week #12: July 7 – 11, 2025	AREA ASSIGNMENT	Software Development
TASK	Project Development	SHIFT/TIME	8:00 AM – 5:00 PM

SOFTWARE DEVELOPMENT:

Starting for this week I added a new feature that required modification to our database and backend. I noticed this feature can not be implemented without that modification also it is crucial as I believe this is one of the fundamental feature or functionalities for our assigned section of the project. As the other teams finished their task or parts of the project, we begun the merging our team's work into the main project. For this we used the GitHub pushing our API's Code on the remote main repository. After this we transition from localhost to Public API that was given by our supervisor, then we tested our section using Insomnia and the Mobile GUI we developed.

TECHNICAL DOCUMENTATION AND SKILLS DEVELOPMENT:

Our Team collaborated with the other teams to merge the Project into one via MS Team video call to make sure that there will be no problem and answer all inquiries immediately. Making the work easier and efficient. As we tested them on our Personal computer we were tasked for each team to create a explanatory video regarding the application. Adding the feature become a little challenging since I also need to change the backend a little but the backend is already merge in the main branch. Despite, the challenges we were able to finish it within the day.



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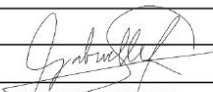
DATE	Week #13: July 14 – 18, 2025	AREA ASSIGNMENT	Software Development
TASK	Project Development	SHIFT/TIME	8:00 AM – 5:00 PM

SOFTWARE DEVELOPMENT:

For this week our team focused more on merging the modules. With our Module 3 Section we connected it with Module 0 and 1 database since we are using a temporary table or collection and a hardcoded user Id. Connecting the Module 3 with Module 0 and 1 which the account authorization will come we were able to get it to integrate with the module 3 section and remove the hardcoded that are needed. With these merging its inevitable to do a drastic change in name variable and variable type. I also created a video for our module, a demo of our module submitting it on our supervisor.

TECHNICAL DOCUMENTATION AND SKILLS DEVELOPMENT:

In here I were able to understand clearly the merging and integrating with another module better. Carefulness is needed to not mess with other code or other module and also the communication between out team increases because of this.



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DATE	Week #14: July 21 – 25, 2025	AREA ASSIGNMENT	Software Development
TASK	Project Development	SHIFT/TIME	8:00 AM – 5:00 PM

SOFTWARE DEVELOPMENT:

This is the last week, so we were just cleaning up and doing some final work. We fixed some modules of ours in Module 3 because ours were accidentally affected when another module was merged. Fortunately, it was eventually fixed and no problem arose afterwards. We also created a recording this time with a more complete module of our section, and it was sent to our supervisor for confirmation.

TECHNICAL DOCUMENTATION AND SKILLS DEVELOPMENT:

Since we used GitHub a lot for this project, we had to merge the modules and the collaboration of the groups and teams. I was able to enhance my skills in collaboration also acquire better skills in GitHub.



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